

Total No. of Questions : 8]

[Total No. of Printed Pages : 1

Roll No

EC-7011 (GS)

B.E. VII Semester Examination, June 2020

Grading System (GS)

Wireless Communication

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. Consider an indoor wireless LAN with frequency 900MHz, cells of radius 100m and nondirectional antenna. Under the free space path loss model, what transmit power is required at the access point such that all terminals within the cell receives a minimum power of $10\mu\text{W}$. How does this change in the system frequency is 5GHz?
2. How propagation is achieved in view of diffraction and scattering by rough services? Explain.
3. Discuss the characterization of deterministic linear time variant systems.
4. With suitable example explain Mathematical Model of Narrowband and wideband channels.
5. Explain the principle of combining diversity. What are the different techniques used for combining diversity?

OR

What do you mean by blind equalizer? Why it is used in wireless communication? Explain in detail?

6. Find the critical distance d_c under two path model for a large microcell in a suburban area with the base station mounted on a tower of height 30 m, the receiving at height 6 m and frequency -2.5 GHz. Is this a good size for cell radius in a suburban microcell? Why or why not?
7. Write a brief note on the antennas used for mobile stations.
8. Answer any two of the following :
 - a) Differentiate between regulated and unregulated spectrum in frequency reuse.
 - b) Compute the Rayleigh distance of a square antenna with 20 dB gain.
 - c) Differentiate between Inverse filtering and averaging used in equalizer.
 - d) What is Power-spectral density of cyclostationary processes.
